



DATA-AI-EXPERIENCE (DAX) CHALLENGE 2026 TECHNICAL REPORT

NO WATER, NO LIFE: Uncovering Pahang's Water Supply Crisis

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INTRODUCTION

1.1 Research Background

Water is one of the most essential resources for sustaining households, agriculture, industries, and economic development. A reliable water supply is crucial to ensure the well-being of communities and support continuous socioeconomic growth. However, water supply systems are increasingly challenged by fluctuating water demand, ageing infrastructure, pipe failures, environmental conditions, and operational limitations, which may result in low water pressure and supply disruptions across different districts.

The importance of addressing water supply issues in Pahang is highlighted by a real disruption event reported during the El Niño phenomenon in 2016. According to Bernama through Astro Awani, more than 7,000 water account holders in Temerloh, Lipis, and Pekan experienced water supply disruptions due to low water pressure at several water treatment plants, with Lipis being the most severely affected district (Bernama, 2016). This incident demonstrated how extreme environmental conditions can significantly affect water supply reliability and disrupt the daily lives of thousands of residents.

In 2022, the Pahang State Government announced a long-term commitment to resolve at least 90% of water supply problems within five years through extensive infrastructure and water supply improvement projects in order to recognise the importance of improving water services (Bernama, 2022). The initiative involved major investments in projects addressing supply disruptions, low water pressure, burst pipes, water distribution networks, and treatment plant efficiency, reflecting the government's commitment to providing a more reliable water supply for the people of Pahang.

Although infrastructure development plays an important role in solving water supply problems, effective decision-making also requires continuous monitoring and analysis of historical water supply data. Water demand, production capacity, non-revenue water (NRW), water losses, customer consumption, infrastructure performance, and supply disruption patterns can provide valuable insights into districts experiencing greater pressure within the water supply system. Analysing these indicators together enables stakeholders to identify trends, compare district performance, and recognise potential areas requiring earlier attention.

Therefore, this project aims to develop a Pahang Water Supply Analytics Dashboard using PAIP historical data to provide a comprehensive and interactive analysis of water supply conditions across districts. The dashboard visualises key indicators including total water demand, water production, capacity utilisation, non-revenue water, water loss ratio, per capita consumption, and water supply disruption hours. By transforming historical operational data into meaningful visual insights, this project seeks to support evidence-based monitoring and assist government agencies and water authorities in identifying districts with higher water supply risks,

prioritising resource allocation, and contributing towards the broader government objective of reducing water supply problems throughout Pahang.

1.2 Problem Statement

Water supply disruptions remain a significant challenge in Pahang, affecting residents, businesses, and other water users. The problem is influenced by the interaction of increasing water demand, infrastructure conditions, water losses, environmental factors, and operational disruptions. However, the large number of water-related indicators makes it difficult to determine which districts are most vulnerable, when demand pressure is highest, and which factors are most closely associated with water supply disruptions.

Currently, utility managers lack diagnostic capabilities to systematically isolate districts experiencing erratic, unexplained consumption spikes versus normal seasonal growth. This diagnostic limitation hinders their ability to detect localized metering errors, illegal tapping, or unbilled commercial surges. Furthermore, operational and financial indicators such as capacity utilization, infrastructure aging, average plant age of 23.1 years and meter age of 9.3 years, hydraulic pressure drops, pipe burst frequency, and NRW percentage are currently evaluated in isolation.

Therefore, this study investigates the patterns and factors associated with water supply disruptions across Pahang. The study develops analytical indicators, including a Water Crisis Score and crisis-level classification, to diagnose the severity of water supply conditions at the district level. In addition, Power BI-based analytical dashboard and forecasting techniques are used to identify consumption patterns and forecast Water Crisis Scores for the following 12 months. The findings aim to provide data-driven insights that can help water authorities identify vulnerable districts, anticipate potential future risks, and improve the prioritisation of water resource management and intervention strategies.

1.3 Research Question

- What are the key factors associated with water supply disruptions across the districts in Pahang?
- What are the water consumption patterns in Pahang in terms of consumption sector, geographical distribution, and seasonal trends?
- How to develop an interactive Power BI dashboard that provides data-driven insights to support the monitoring, prioritisation, and management of water supply problems in Pahang?

1.4 Research Objectives

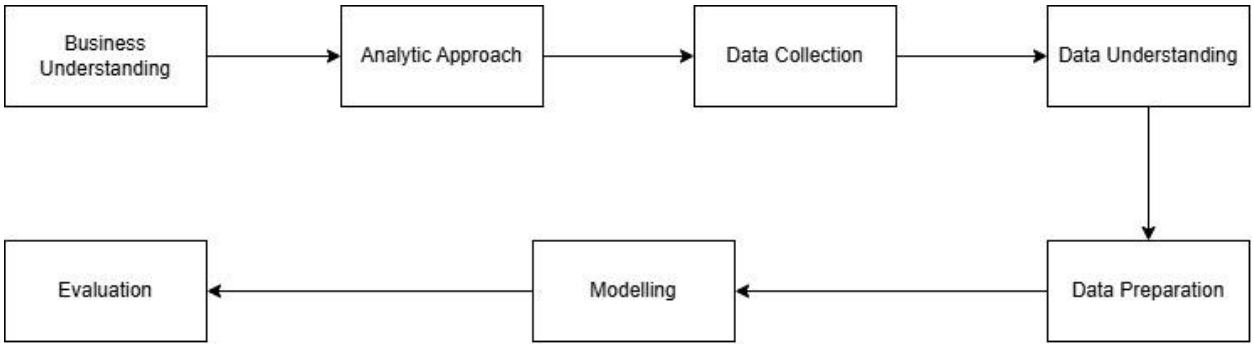
- To identify the key factors associated with water supply disruptions across the districts in Pahang
- To analyze the water consumption patterns in Pahang in terms of consumption sector, geographical distribution, and seasonal trends
- To develop an interactive Power BI dashboard that provides data-driven insights to support the monitoring, prioritisation, and management of water supply problems in Pahang.

1.5 Significance of Research

This dashboard is significant to Pengurusan Air Pahang Berhad (PAIP) and the Pahang state government to solve Pahang's water supply disruptions issues by providing recommendation action. By consolidating factors such as supply disruption hours, customer complaints, non-revenue water (NRW) loss, and pipe bursts into a single Water Crisis Score, PAIP can systematically identify critical distinct and target capital expenditure directly where network strain is highest.Additionally, the dashboard's risk forecasting enables Pahang utility managers to predict seasonal outage spikes before they occur. This early warning system allows PAIP to perform preventative maintenance, deploy emergency water supplies, and optimize resource allocation in advance. Ultimately, this tool directly supports Pahang's state initiative to resolve at least 90% of local water supply issues within five years while advancing UN Sustainable Development Goals of SDG 6, 11, and 12 for resilient water infrastructure across the state.

METHODOLOGY

The dataset of this study will be processed following this data science methodology cycle.



2.1 Business Understanding

This study focuses on investigating and managing water supply disruptions across districts in Pahang by examining the condition of the water supply system from multiple perspectives. The analysis focuses on five key areas: diagnosing the severity of water crises, identifying factors contributing to disruptions, understanding water consumption patterns, predicting future water crisis risks, and recommending appropriate actions. By integrating these aspects, the study aims to provide a comprehensive understanding of the current and potential future water supply situation in Pahang.

Therefore, the main business goal of this study is to develop a data-driven decision-support system for investigating and managing water supply disruptions across Pahang. The analysis uses PAIP's water supply data from 2023 to 2025 to assess the condition of the water supply system across different districts. A Water Crisis Score is developed to diagnose and compare the severity of water supply pressure among districts. The analysis then investigates the potential factors associated with disruptions, including infrastructure conditions, pipe burst incidents, raw water turbidity, rainfall, and other operational indicators.

In addition, the study examines the water consumption profile to determine who contributes most to water demand, where demand is concentrated, and when seasonal consumption peaks occur. Understanding these consumption patterns is important for identifying periods when the water supply system may experience greater pressure. To move from historical analysis towards proactive planning, the Water Crisis Score is also forecast for the following 12 months for each district. This allows potential future increases in water crisis risk to be identified before they become more severe.

As a result, the findings from this study can assist PAIP and relevant state government authorities in making more informed decisions regarding water supply management. The results can help authorities identify high-risk districts, understand potential causes of disruptions, anticipate future crisis levels, prioritise infrastructure maintenance and resources, and implement targeted preventive actions.

2.2 Analytic Approach

To evaluate water supply stability across Pahang, this study implements a four-stage analytical framework integrating descriptive, diagnostic, consumption, and predictive methodologies. The descriptive phase utilizes PAIP operational records from 2023 to 2025 across 11 districts to establish a baseline performance index. Key to this stage is the composite Water Crisis Score, which combines four normalized metrics: supply disruption hours (40%), pipe burst incidents (20%), customer complaints (20%), and Non-Revenue Water (NRW) percentages (20%). The final numerical score categorizes each district into four distinct risk tiers which are Low, Moderate, High, and Critical, which enables comparison across the state.

The diagnostic phase investigates the underlying causes of supply disruptions by focusing on two main areas: structural failure and weather impacts. Structural risks are evaluated through localized pipe burst density (incidents per kilometer) across urban (Bandar), rural (Luar Bandar), and suburban (Pinggir Bandar) zones, alongside operational limits of aging treatment plants. Weather risks are examined by tracking how heavy rainfall causes river water turbidity to surge above 300–500 NTU, which clogs legacy filtration systems and triggers plant shutdowns.

The consumption profile analysis examines usage patterns across domestic, commercial, and industrial customer groups. By analyzing daily demand cycles and peak usage periods, this stage highlights when high customer demand overlaps with existing physical network weaknesses, further increasing system pressure and risk of failure.

The predictive phase projects water crisis risks 12 months forward through December 2026. Using an Exponential Smoothing (ETS) time-series model alongside DAX-based linear regression (LINESTX), the study forecasts future district-level Water Crisis Scores and calculates the Predicted Risk Percentage Change relative to current baselines. This quantitative metric measures the expected growth or decline in system stress, allowing authorities to identify emerging crisis hotspots in advance.

2.3 Data Collection

The primary dataset for this study was collected directly from Pengurusan Air Pahang Berhad (PAIP) including the years from 2023 to 2025. This dataset contains 2,664 monthly observations across 50 variables. To translate the variables into actionable insights, the variables are composited to Water Crisis Score, Water Crisis Level, Burst Density, Predicted Risk Percentage Change and Forecast Crisis Score 2026. These composite variables summarize complex operational data into clear risk indicators and forward-looking forecasts, enabling PAIP to quickly identify infrastructure vulnerabilities and proactively manage regional water security.

2.4 Data Preparation

An initial null-value assessment revealed that the vast majority of the dataset was complete. However, missing values were isolated in two specific environmental metrics. Raw Water Turbidity (NTU) contained 89 missing entries, which were imputed using the median turbidity value grouped by the respective district to account for regional environmental similarities, alongside an overall median fallback. Additionally, Average Pressure (bar) contained 58 missing values. Since pressure is highly localized to specific infrastructure, these entries were imputed using the median pressure grouped by the specific plant name, again utilizing an overall median fallback to ensure complete data integrity.

2.5 Data Understanding

The dataset used in this study was provided by the Pahang Water Supply Department (PAIP) and consists of 50 variables containing information related to water production, consumption, infrastructure, water losses, customer demand, environmental conditions, and water supply disruptions across districts in Pahang. The dataset covers a three-year period from 2023 to 2025, allowing historical trends and patterns in water supply conditions to be examined over time. These variables provide a comprehensive view of the state's water supply conditions and allow different aspects of water supply performance to be evaluated across districts and years.

2.6 Data Modelling

The data modelling phase transforms raw utility variables into a structured analytical framework to evaluate current water supply conditions and predict future infrastructure risks. This process bridges historical data assessment with advanced forecasting techniques to generate actionable insights.

2.6.1 Dax Measures

Several DAX measures were created in Power BI to derive additional indicators from the original variables. These measures were developed to support the analysis and provide more

meaningful metrics for evaluating water supply conditions, such as water demand, water losses, revenue loss, and water crisis levels. The created DAX measures are presented in the table below.

No	Dax Measure	Purpose	Formula
1	Water Crisis Score	Calculates a composite numerical index to measure the severity of water stress by weighting key operational metrics (supply disruptions, pipe bursts, complaints, and NRW percentage).	Water Crisis Score = VAR DisruptionScore = DIVIDE(AVERAGE('PAIP'[Supply_Disruption_Hours]), CALCULATE(AVERAGE('PAIP'[Supply_Disruption_Hours]), ALL('PAIP'[District]))) VAR BurstScore = DIVIDE(AVERAGE('PAIP'[Pipe_Burst_Incidents]), CALCULATE(AVERAGE('PAIP'[Pipe_Burst_Incidents]), ALL('PAIP'[District]))) VAR ComplaintScore = DIVIDE(AVERAGE('PAIP'[Customer_Complaints]), CALCULATE(AVERAGE('PAIP'[Customer_Complaints]), ALL('PAIP'[District]))) VAR NRWScore = DIVIDE(AVERAGE('PAIP'[NRW_Percentage]),

			CALCULATE(AVERAGE('PAIP'[NRW_Percentage]), ALL('PAIP'[District]))) RETURN (DisruptionScore * 0.40 + BurstScore * 0.20 + ComplaintScore * 0.20 + NRWScore * 0.20)
2	Water Crisis Level	Categorizes the numerical system strain or crisis score into clear, qualitative diagnostic tiers (e.g., Normal, Watch, Critical) for immediate visual assessment on the dashboard.	Water Crisis Level = VAR Score = [Water Crisis Score] RETURN SWITCH(TRUE(), Score < 0.75, "🟢Low Level of Water Crisis", Score < 1.00, "🟡Moderate Level of Water Crisis", Score < 1.25, "🔴High Level of Water Crisis", "⚠️Critical Level of Water Crisis")
3	Burst Density	Calculates the frequency of pipe failures relative to the size of the infrastructure network (bursts per kilometer).	Burst_Density = DIVIDE(SUM(PAIP[Pipe_Burst_Incidents]), SUM(PAIP[Pipe_Length_km]), 0)
4	Predicted Risk Percentage Change	Calculates the projected percentage growth or decline in	Predicted Risk Percentage Change = VAR Baseline = [Current Crisis Score] VAR ForecastVal = [Forecast Crisis Score

		the Water Crisis Score between the current baseline and the end of 2026.	2026] RETURN IF(ISBLANK(Baseline) || ISBLANK(ForecastVal), BLANK(), DIVIDE(ForecastVal - Baseline, Baseline, 0))
5	Forecast Crisis Score 2026	Predicts the expected Water Crisis Score based on the overarching historical trend of the dataset using linear regression method.	Forecast Crisis Score 2026 = VAR BaseDate = CALCULATE(MIN('PAIP'[Date]), REMOVEFILTERS('PAIP'[Date])) VAR TargetDate = DATE(2026, 12, 31) VAR DataPoints = ADDCOLUMNS(FILTER(ALL('PAIP'[Date]), NOT ISBLANK([Water Crisis Score])), "X", 'PAIP'[Date] - BaseDate, "Y", [Water Crisis Score]) VAR RegTable = LINESTX(DataPoints, [Y], [X]) VAR Slope = SELECTCOLUMNS(RegTable, [Slope1]) VAR Intercept = SELECTCOLUMNS(RegTable, [Intercept]) VAR TargetX = TargetDate - BaseDate RETURN Intercept + (Slope * TargetX)

2.6.2 Predictive Modelling

To extend the analysis from historical assessment to future risk prediction, an Exponential Smoothing (ETS) forecasting model was applied to the historical Water Crisis Score. ETS is a time-series forecasting approach that considers three main components, Error (E), Trend (T), and Seasonality (S). The model gives greater importance to recent observations while using historical patterns to estimate future values. This makes ETS suitable for analysing the Water Crisis Score, which may contain changes in level, trends, and recurring seasonal patterns.

In this study, the historical Water Crisis Score from 2023 to 2025 was used to forecast the expected crisis score for the next 12 months in 2026. The model analyses the historical time-series pattern for each district and generates future Water Crisis Score predictions. A 95% confidence interval is also generated to represent the expected range of future values and indicate the uncertainty surrounding the predictions. Power BI's forecasting functionality supports forecasting future values from historical time-series data and allows the forecast length and confidence interval to be specified.

The forecasted score is then compared with the latest observed Water Crisis Score to calculate the Predicted Risk Percentage Change. A positive percentage change indicates that the district's water crisis risk is expected to increase, while a negative percentage change indicates a potential improvement. The results allow high-risk districts and potential future increases in water crisis levels to be identified, supporting proactive maintenance, resource allocation, and water supply planning.

2.7 Data Evaluation

To establish a rigorous and accurate baseline for evaluating the water supply network, several customized statistical formulas and threshold-based approaches were developed to process and interpret the raw PAIP data. First, a Composite Crisis Score was formulated as a weighted statistical index to quantify the level of water supply pressure across different districts. The index normalizes historical values and assigns specific weights to key operational indicators such as Supply Disruption Hours, customer complaints, NRW percentage, and pipe burst incidents. This approach ensures that indicators with greater relevance to water supply disruptions have an appropriate influence on the overall crisis score. The table below shows the indicators and their corresponding weightage contributed for the composite crisis score.

Indicators	Weightage (%)
Supply Disruption Hours	40
Customer Complaints	20
NRW Percentage	20
Pipe Burst Incidents	20

In addition, Crisis Level Thresholds were established to convert the continuous Water Crisis Score into several qualitative categories. This classification simplifies the interpretation of numerical results and allows districts with higher levels of water supply vulnerability to be identified more easily. The category level of water crisis and the interpretations is shown as table below:

Water Crisis Score	Interpretation
0 ~ 0.74	Low Level of Water Crisis
0.75 ~ 0.99	Moderate Level of Water Crisis
1.0 ~ 1.24	High Level of Water Crisis
1.25 above	Critical Level of Water Crisis

RESULTS AND DISCUSSIONS

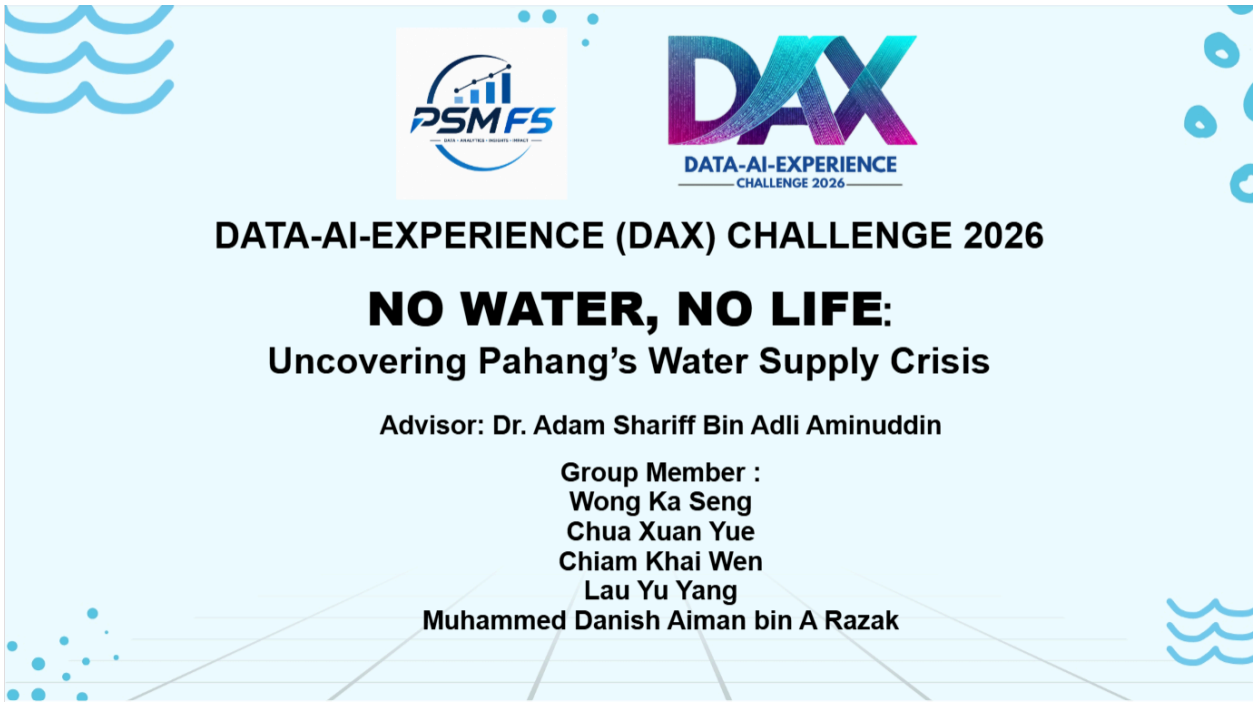
3.1 Dashboard

This section presents four interconnected Power BI dashboards developed to address the water supply crisis in Pahang, each covering a distinct stage of the investigation. The first dashboard establishes the diagnostic baseline by ranking districts according to their Water Crisis Score, revealing where disruptions are most severe. The second dashboard moves from what is happening to why, examining structural and environmental root causes such as pipe burst density, treatment plant age, and rainfall-driven turbidity.

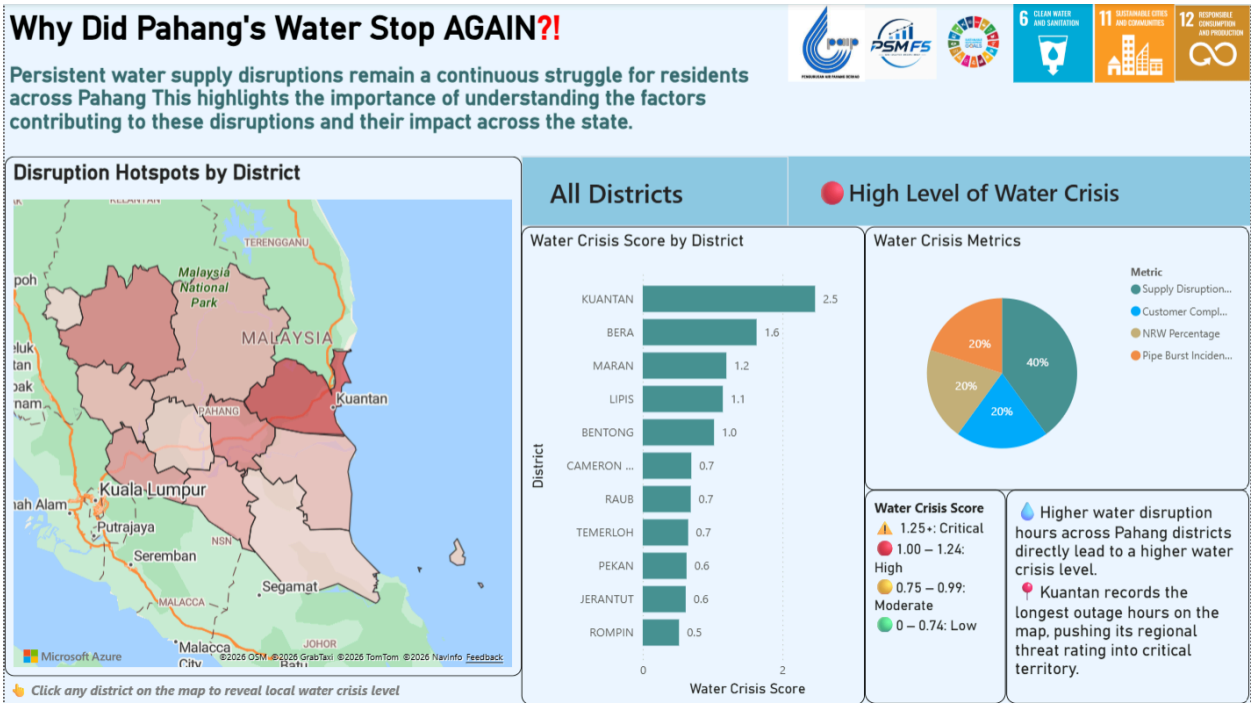
Next, the third dashboard shifts focus to demand-side dynamics, illustrating how consumption patterns across sectors and seasons place additional pressure on the supply system. Finally, the fourth dashboard looks ahead, using Power BI's forecasting capabilities to project Water Crisis Scores through 2026 and flag districts at risk of worsening conditions.

The combination of all dashboards guide the audience through a natural progression, starting with diagnosing the problem, then uncovering its root causes, understanding what's driving demand, and finally anticipating what may come next, giving PAIP and state authorities a complete, evidence-based view for prioritising interventions.

Cover Page: NO WATER, NO LIFE: Uncovering Pahang's Water Supply Crisis



Dashboard 1: Why Did Pahang's Water Stop AGAIN?!



Interpretation:

The dashboard illustrates the overall severity and spatial distribution of water supply disruptions across Pahang districts supporting the focus on SDG 6 Clean Water and Sanitation, SDG 11 Sustainable Cities and Communities and SDG 12 Responsible Consumption and Production. The findings highlight significant regional disparities in water crisis levels, driven primarily by long outage hours, customer complaints, Non-Revenue Water (NRW) challenges and pipe burst incidents.

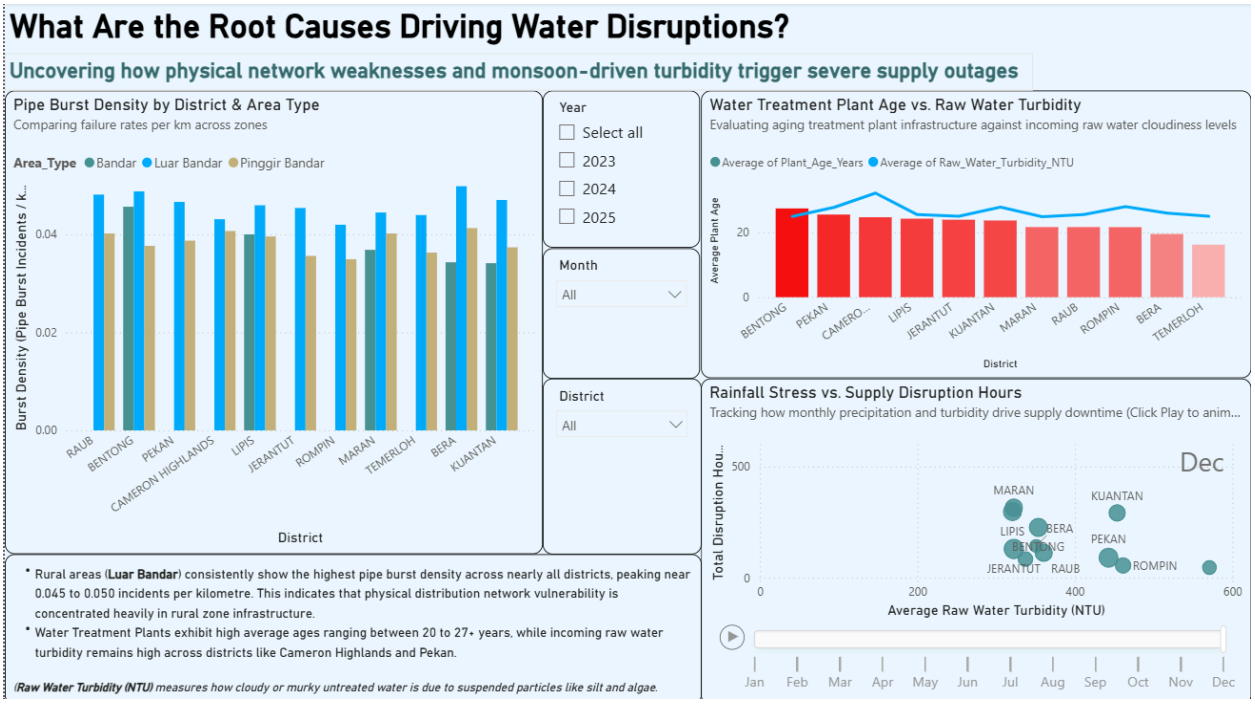
First, the horizontal bar chart ranks districts by their overall Water Crisis Score. Kuantan records the highest crisis score of 2.5, significantly higher than Bera (1.6) and Maran (1.2). This underlines Kuantan has a critical level of water crisis and it is the primary disruption hotspot in the state, where supply interruptions occur most frequently and severely.

Second, the pie chart breaks down the core metrics contributing to the water crisis rating. Supply disruption hours account for the largest proportion at 40%, while customer complaints, NRW percentage, and pipe burst incidents each contribute 20%. Hence, higher water supply disruption hours will directly lead to a higher water crisis level.

Lastly, the map visualizes these disruption hotspots geographically across Pahang. Districts shaded in darker red, particularly Kuantan, correspond to higher crisis ratings driven by

prolonged outage durations that push their regional threat status into critical territory. The map also allows users to reveal the district's water crisis level by clicking the specific district.

Dashboard 2 : What Are the Root Causes Driving Water Disruptions?



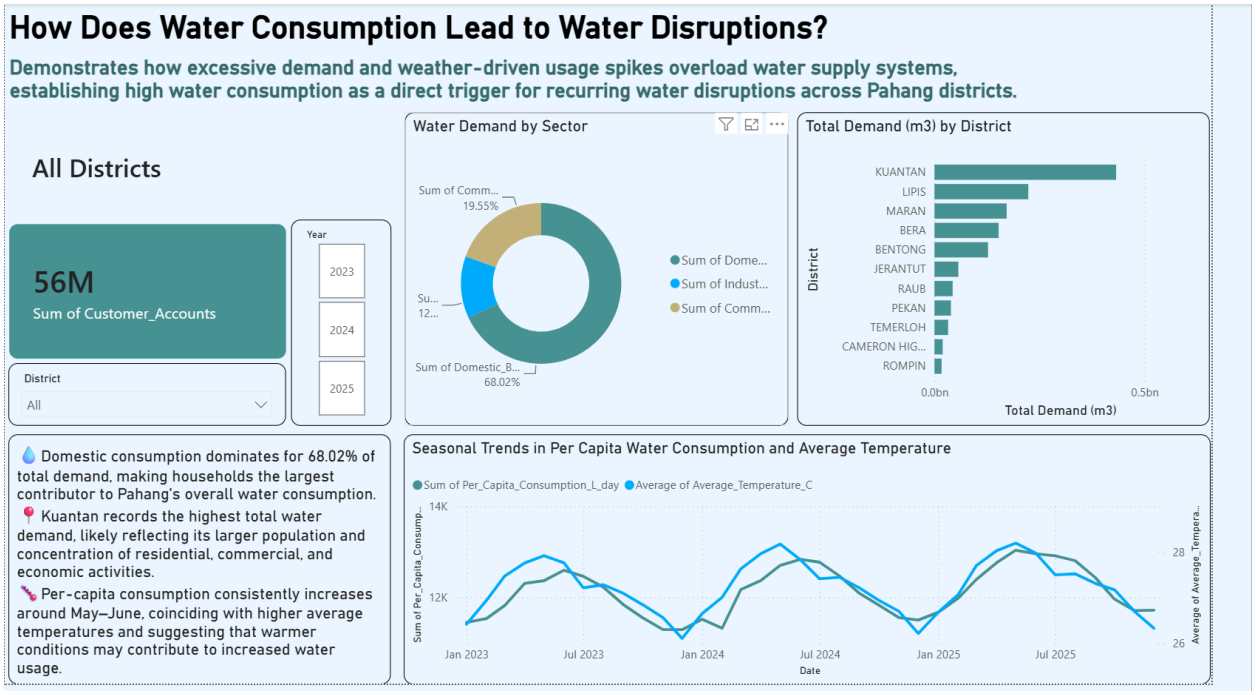
Interpretation:

The dashboard investigates the structural and environmental root causes driving water supply disruptions across districts in Pahang. Rather than measuring operational pressure, the visual analysis focuses on physical distribution vulnerabilities, water treatment plant longevity, and the operational impact of monsoon-driven turbidity spikes.

Analyzing physical distribution networks, the pipe burst density chart demonstrates that rural zones (*Luar Bandar*) consistently suffer the highest failure rates per kilometer in every district.. This indicates that structural distribution vulnerability is heavily concentrated in rural infrastructure compared to urban (*Bandar*) and suburban (*Pinggir Bandar*) areas.

Simultaneously, the performance of treatment infrastructure is constrained by plant age and incoming water quality. Water Treatment Plants across Pahang exhibit high average ages ranging from 18 to over 27 years, with older facilities in districts like Bentong and Pekan frequently subjected to elevated raw water turbidity levels. During peak monsoon months such as December, incoming raw water turbidity surges up to 600 NTU, resulting in severe supply downtime that exceeds 300 disruption hours in heavily impacted districts like Kuantan and Maran.

Dashboard 3: How Does Water Consumption Lead to Water Disruptions?



Interpretation:

The dashboard illustrates the relationship between water consumption patterns, customer sectors, district-level demand, and seasonal conditions in Pahang from 2023 to 2025. Overall, the findings indicate that water demand is strongly concentrated in certain customer sectors and districts, while consumption tends to increase during warmer months.

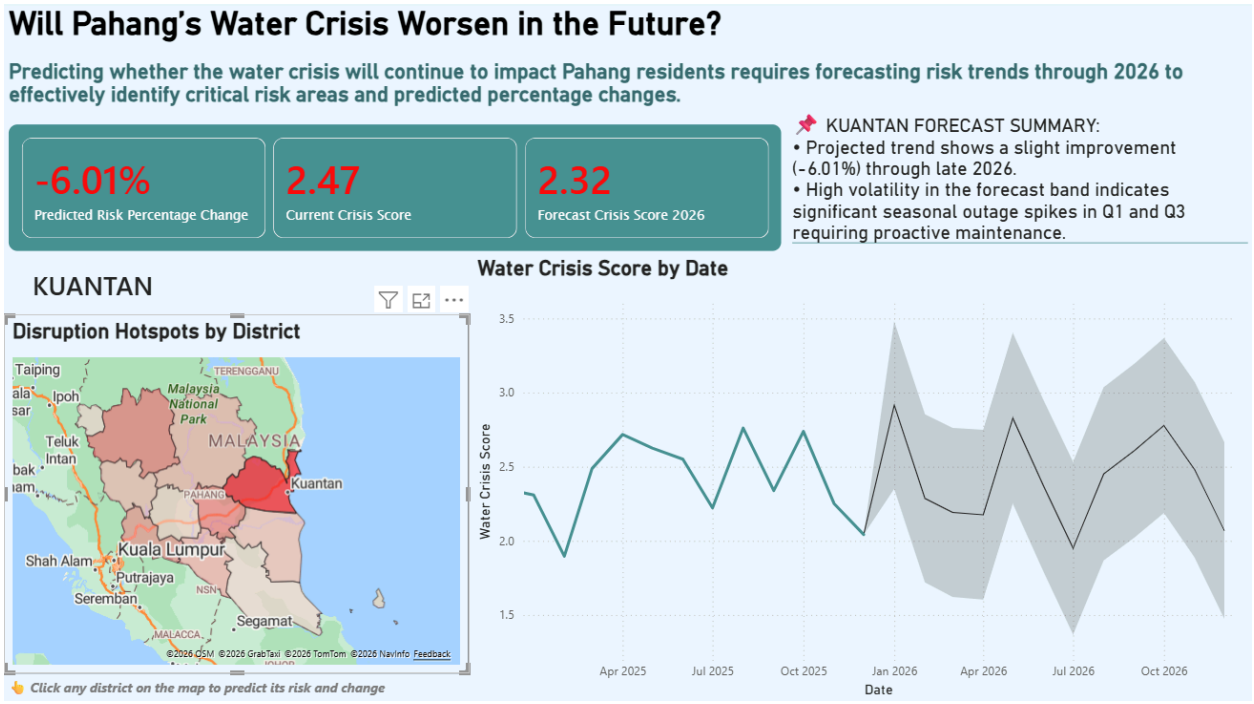
Primarily, the donut chart shows that the domestic sector recorded the highest water consumption at 68.02%, followed by the commercial sector at 19.55% and the industrial sector at 12.43%. The high contribution from domestic consumption reflects the large number of residential users and their continuous daily water requirements.

Secondly, a clustered bar chart is used to show total water demand at different district levels. Kuantan recorded the highest total water demand, which is likely associated with its larger population and concentration of residential, commercial, and economic activities. Kuantan has the highest sum of customer accounts compared to other districts, which leads to greater water requirements.

Moreover, the line chart shows the seasonal trend further shows recurring peaks in per-capita water consumption around May to June, coinciding with higher average temperatures of approximately 28°C or above. Warmer conditions may increase household water usage, resulting in higher overall demand during this period. These recurring consumption peaks

highlight the importance of monitoring seasonal demand and ensuring sufficient water supply capacity during periods of higher consumption.

Dashboard 4 : Will Pahang's Water Crisis Worsen in the Future?



Interpretation:

The dashboard evaluates predictive risk trends through late 2026 to determine whether water disruptions in Pahang will escalate, focusing specifically on forecasted crisis scores for individual districts such as Kuantan.

Analyzing the overall outlook for Kuantan, the forecast indicates a minor decrease in overall severity, reflected by a -6.01% Predicted Risk Percentage Change as the district's crisis score drops from a current baseline of 2.47 to a projected score of 2.32 in 2026. Spatial mapping shows Kuantan highlighted in bright red on the map, designating it as a critical disruption hotspot relative to neighboring districts.

Power BI uses a forecasting method called ETS AAA, which stands for error, trend, and seasonality when all treated in an additive way. This method is a type of exponential smoothing, meaning it predicts future values by giving more weight to recent data points and gradually less weight to older ones, instead of treating every past data point equally.

Despite the slight long-term reduction in average risk score, the time-series forecasting model reveals severe seasonal volatility within the shaded confidence interval band. The

projection trajectory identifies major quarterly outage spikes most prominently in Q1 and Q3 2026 demonstrating that seasonal disruptions will remain a significant operational challenge requiring targeted proactive maintenance.

Dashboard 5: Recommended Actions



Based on our forecasting results, we recommend five key strategies to tackle Pahang's water crisis and move from constant firefighting to proactive planning. These evidence-based actions directly support UN Sustainable Development Goals 6, 11, and 12, building a more resilient and reliable water supply network across the state.

The first is deploying IoT burst systems to monitor pipe conditions in real time. Installing pressure and acoustic sensors in busy areas like Kuantan and spread-out rural zones gives utility teams continuous visibility over the network. Pressure sensors catch sudden drops that signal major pipe breaks, while acoustic sensors pick up the faint vibrations of small leaks before pipes fail completely. Placing these devices in Kuantan handles the heavy urban load, while coverage in rural areas makes up for long physical inspection distances. This real-time data helps crews fix issues quickly, saving millions of liters of lost water and preventing sudden supply cuts.

The second is carrying out pre-peak preventative maintenance timed with seasonal weather patterns. Our time-series forecasting model revealed clear risk spikes happening every year during Q1 and Q3. Instead of waiting for machinery to break down during these peak periods, maintenance crews should service pumps, filtration units, and valves every December and May. Tuning up equipment right before high-demand windows ensures treatment plants

operate at full capacity when the system is under the heaviest stress, preventing costly breakdowns during crisis months.

The third is targeted pipe replacement and active leak detection using smart capital investment. A major source of water loss in Pahang comes from legacy distribution mains that are over 20 years old, making them brittle and prone to bursting. Rather than spreading the budget thinly across the state, funding should focus on replacing these high-risk aging pipes while using ground listening equipment to hunt for hidden subterranean leaks. Upgrading these old sections stabilizes water pressure, cuts down unrecorded physical water loss, and extends the overall life of the network.

The fourth is establishing a dedicated water supply buffer in Kuantan to handle sudden outages. Because Kuantan has the highest population density and water demand in Pahang, any main break creates an immediate crisis for thousands of residents. Installing permanent static storage tanks and keeping mobile water tankers ready in high-risk neighborhoods creates a reliable backup system. If a main pipe bursts or a treatment plant shuts down unexpectedly, these stored reserves keep clean water flowing to homes and critical facilities without delay.

The fifth is implementing an early weather response protocol to protect water supplies against climate shifts. Water utility operators need to fill raw water storage reservoirs well ahead of predicted monsoons or dry spells. Filling reservoirs before the monsoon hits avoids drawing in muddy, debris-filled river water during heavy storms, which clogs treatment plant filters. Similarly, storing extra water before forecasted droughts guarantees adequate stock before river levels drop too low, keeping clean drinking water flowing through extreme weather changes.

3.2 Recommendations & Potential State Impact

By connecting the dashboard findings with practical field actions, these five recommendations give Pahang a clear roadmap to fix its water crisis at the root. They turn predictive insights into real-world fixes while directly supporting United Nations Sustainable Development Goals for clean water (SDG 6), sustainable communities (SDG 11), and responsible resource management (SDG 12).

The main goal is to stop physical water leaks and protect state revenue. Setting up IoT pressure and sound sensors across Kuantan and rural areas targets the high pipe burst zones flagged in Dashboard 2 (SDG 11), catching micro-leaks and pressure drops before main lines break. Pairing this technology with targeted replacements for pipes over 20 years old directly tackles Pahang's massive 40.74% Non-Revenue Water (NRW) rate highlighted in Dashboard 1 (SDG 6). Stopping this physical wastage keeps water moving at steady pressure while preventing wasted water resources and lost utility revenue (SDG 12).

To keep plant equipment running smoothly and avoid unnecessary equipment failure (SDG 12), servicing schedules should be timed for May and December, right before the heavy Q1 and Q3 demand spikes shown in Dashboards 3 and 4. In Kuantan, where outages hit hardest, setting up static storage tanks and backup water tankers provides an essential safety net during sudden breakdowns (SDG 11). Combining this local buffer with an early weather response protocol, filling raw water reservoirs ahead of monsoons or dry spells (SDG 6) which prevents muddy river water and droughts from shutting down treatment plants. Together, these steps directly target Pahang's historical 20,150.1 disruption hours, ensuring steady, reliable clean water for over 7,000 account holders across the state.

Conclusion

This study successfully investigated water supply conditions and disruptions across the districts of Pahang using PAIP data from 2023 to 2025. The analysis addressed the research questions by examining the factors associated with water supply disruptions, identifying water consumption patterns, and developing an interactive Power BI dashboard to integrate the findings into a practical decision-support tool.

The analysis had identified several important operational, infrastructure, and environmental indicators associated with water supply conditions. Factors such as pipe burst incidents, customer complaints, non-revenue water, production, capacity utilization, rainfall, temperature, and raw water turbidity provided insights into the different conditions that may contribute to water supply pressure and disruptions across districts. The Water Crisis Score further enabled the severity of water supply conditions to be compared between districts.

Moreover, the findings showed that the domestic sector was the largest contributor to water consumption, accounting for 68.02%, followed by the commercial and industrial sectors. Kuantan recorded the highest total water demand, which may be associated with its larger population and concentration of residential, commercial, and economic activities. The analysis also identified a recurring increase in per-capita water consumption around May to June, coinciding with higher temperatures. This suggests that seasonal weather conditions may be associated with increased water consumption during this period.

Furthermore, the study successfully developed five interconnected dashboards covering the Water Crisis Score, contributing factors, consumption profile, future Water Crisis Score prediction, and recommended actions. These dashboards transform the PAIP dataset into interactive visual insights that allow users to identify high-risk districts, investigate potential contributing factors, understand consumption patterns, and assess future water crisis conditions.

Overall, the study demonstrates how data analytics, visualization, and predictive forecasting can be integrated to support more proactive water supply management. Therefore, the findings will assist PAIP and relevant government authorities in monitoring water supply conditions, prioritising districts requiring attention, planning resources, and implementing targeted actions.

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